

1. Name of the medicinal product

DexaMedica Meropenem 0.5g Powder for Injection

DexaMedica Meropenem 1g Powder for Injection

2. Qualitative and quantitative composition

DexaMedica Meropenem 0.5g Powder for Injection contains Meropenem trihydrate equivalent to 0.5 g meropenem.

DexaMedica Meropenem 1g Powder for Injection contains Meropenem trihydrate equivalent to 1 g meropenem.

3. Pharmaceutical form

Before reconstitution:

Yellowish white powder

After reconstitution:

The solid dissolved completely. The constituted solution is not significantly less clear than an equal volume of the diluent or of purified water contained in a similar vessel and examined similarly. The solution is essentially free from particulate of foreign matter that can be observed in visual inspection

4. Clinical particulars

4.1 Therapeutic indications

Indicated for treatment, in adults and children, of the following infections caused by single or multiple bacteria sensitive to meropenem.

- Pneumonias and Nosocomial pneumonias
- Urinary Tract Infections
- Intra-abdominal Infections
- Gynaecological Infections, such as endometritis and pelvic inflammatory disease.
- Bacterial Meningitis
- Septicaemia
- Empiric treatment, for presumed infections in patients with febrile neutropenia, used as monotherapy or in combination with anti-viral or anti-fungal agents.

Meropenem has proved efficacious alone or in combination with other antimicrobial agents in the treatment of polymicrobial infections.

4.2 Posology and method of administration

Adults

The dosage and duration of therapy shall be established depending on type and severity of infection and the condition of the patient.

The recommended daily dosage is as follows:

500 mg IV every 8 hours in the treatment of pneumonia, UTI, gynaecological infections such as endometritis.

1 g IV every 8 hours in the treatment of hospital acquired pneumonias, peritonitis, presumed infections in febrile neutropenic patients, septicaemia.

In meningitis the recommended dosage is 2 g every 8 hours.

A dose of up to 2 g three times daily in adults and adolescents and a dose of up to 40 mg/kg three times daily in children may be particularly appropriate when treating some types of infections, such as nosocomial infections due to *Pseudomonas aeruginosa*.

Regular sensitivity testing is recommended when treating *Pseudomonas aeruginosa* infection.

There are limited safety data available to support the administration of a 2 g dose in adults as an intravenous bolus injection.

Special populations

Dosage schedule for adults with impaired renal function

Dosage should be reduced in patients with creatinine clearance less than 51 ml/minute, as scheduled below. There are limited data to support the application of these dose adjustments for a unit dose of 2 g.

Creatinine clearance (ml/minute)	Dose (based on unit doses of 500 mg, 1 g, 2 g)	Frequency
26-50	one unit dose	every 12 hours
10-25	one-half unit dose	every 12 hours
<10	one-half unit dose	every 24 hours

Meropenem is cleared by haemodialysis and haemofiltration; if continued treatment with meropenem is necessary, it is recommended that the unit dose (based on the type and severity of infection) is administered at the completion of the haemodialysis procedure to restore therapeutically effective plasma concentrations.

There is no experience with the use of meropenem in patients under peritoneal dialysis.

Dosage in adults with hepatic insufficiency

No dosage adjustment is necessary in patients with hepatic insufficiency.

Elderly patients

No dosage adjustment is required for the elderly with normal renal function or creatinine clearance values above 50 ml/minute.

Children

For children over 3 months and up to 12 years of age the recommended dose is 10 - 20 mg/kg every 8 hours depending on type and severity of infection, susceptibility of the pathogen and the condition of the patient. In children over 50 kg weight, adult dosage should be used.

In meningitis the recommended dose is 40 mg/kg every 8 hours.

Febrile episodes in neutropenic patients-the dose should be 20 mg/kg every 8 hours. There is no experience in children with renal impairment.

There are limited safety data available to support the administration of a 40 mg/kg dose in children as an intravenous bolus injection.

Route of administration:

Meropenem can be given as an intravenous bolus injection over approximately 5 minutes or by intravenous infusion over approximately 15 to 30 minutes using the specific available presentations.

Meropenem to be used for bolus intravenous injection should be constituted with sterile water for injections (5 ml per 250 mg meropenem). This provides an approximate concentration of 50 mg/ml. The solid dissolved completely. The constituted solution is not significantly less clear than an equal volume of the diluent or of purified water contained in a similar vessel and examined similarly. The solution is essentially free from particulate of foreign matter that can be observed in visual inspection. Meropenem for intravenous infusion may be constituted with compatible infusion fluids (50 to 200 ml).

Meropenem compatibilities with the infusion fluids and the stability after reconstitution are listed below:

Diluent	Stability (hours)	
	25°C	5°C
Water for injection	3	16
0.9% sodium chloride	3	15
5% dextrose infusion	Should be freshly prepared	Should be freshly prepared

Standard aseptic technique should be employed during reconstitution. Shake reconstituted solution before use. It is recommended to use freshly prepared solutions of meropenem for IV injection and infusion.

4.3 Contraindications

Meropenem is contraindicated in patients with known hypersensitivity to the ingredient of drug.

4.4 Warnings and Precautions

- There is some clinical and laboratory evidence of partial cross-allergenicity between other carbapenems and beta-lactam antibiotics, penicillins and cephalosporins. As with all beta-lactam antibiotics, rare hypersensitivity reactions have been reported. Before initiating therapy with meropenem, careful inquiry should be made concerning previous hypersensitivity reactions to beta-lactam antibiotics. Meropenem should be used with caution in patients with such a history. If an allergic reaction to meropenem occurs, the drug should be discontinued and appropriate measures taken.
- As with other antibiotics, overgrowth of nonsusceptible organisms may occur and, therefore, continuous monitoring of each patient is necessary.
- Use of meropenem in patients with hepatic disease should be made with careful monitoring of transaminase and bilirubin levels.
- Use in infections caused by methicillin-resistant staphylococci is not recommended.
- Rarely, pseudomembranous colitis has been reported on meropenem as with practically all antibiotics and may vary in severity from slight to life-threatening. Therefore, antibiotics should be prescribed with care for individual with a history of gastrointestinal complaints, particularly colitis.
- It is important to consider the diagnosis of pseudomembranous colitis in the case of patients who develop diarrhea in association with the use of meropenem. Although studies indicate that a toxin produced by *Clostridium difficile* is one of the main causes of antibiotic-associated colitis, other causes should be considered.
- The co-administration of meropenem with potentially nephrotoxic drugs should be considered with caution.
- Seizures and other CNS adverse experiences have been reported during treatment with meropenem, these experiences have occurred most commonly in patients with CNS disorders (e.g., brain lesions or history of seizures) or with bacterial meningitis and/or compromised renal function.
- Pediatric use: efficacy and tolerability in infants under 3 months old have not been established. Meropenem is not recommended for use below this age. There is no experience in children with altered hepatic or renal function.

4.5 Interaction with other medicinal products and other forms of interactions

- Probenecid competes with meropenem for active tubular secretion and thus inhibits the renal excretion, with the effect of increasing the elimination half-life and plasma concentration of meropenem. As the potency and duration of action of meropenem dosed without probenecid are adequate, the co-administration of probenecid with meropenem is not recommended.
- The potential effect of meropenem on the protein binding of other drugs or metabolism has not been studied. The protein binding of meropenem is low (approximately 2%) and therefore no interactions with other compounds based on displacement from plasma proteins would be expected.
- Meropenem may reduce serum valproic acid levels and anticonvulsant effect. Subtherapeutic levels may be reached in some patients.
- Concurrent use of live typhoid vaccine and antibiotics may result in a decreased immunological response to the typhoid vaccine.

4.6 Pregnancy and lactation

Pregnancy

Pregnancy category B

The safety of meropenem in human pregnancy has not been evaluated. The only adverse effect observed in animal reproductive studies was an increased incidence of abortions in monkeys at 13 times the expected exposure in man. Meropenem should not be used in pregnancy unless the potential benefit justifies the potential risk to the fetus. In every case, it should be used under the direct supervision of the physician.

Lactation

Meropenem is detectable at very low concentrations in animal breast milk.

Meropenem should not be used in breast-feeding women unless the potential benefit justifies the potential risk to the baby.

4.7 Effects on ability to drive and use machines

No studies on the effect on the ability to drive and use machines have been performed.

4.8 Undesirable effects

Serious adverse reactions are rare. The following adverse reactions have been reported:

- Local intravenous injection site reactions: inflammation, thrombophlebitis, pain at the site of injection.
- Skin reactions: rash, pruritus, urticaria. Rarely, severe skin reactions such as erythema multiforme, Stevens-Johnson syndrome and toxic epidermal necrolysis have been observed.
- Systemic allergic reactions: rarely, systemic allergic reactions (hypersensitivity) may occur following administration of meropenem. These reactions may include angioedema and manifestations of anaphylaxis, such as shock, hypotension, respiratory depression.
- Gastrointestinal: abdominal pain, nausea, vomiting, diarrhea. Pseudomembranous colitis has been reported.
- Blood: thrombocytopenia, eosinophilia, thrombocytopenia, leucopenia and neutropenia (including very rare cases of agranulocytosis), haemolytic anemia, and bleeding. A positive direct or indirect Coombs test may develop in some subjects; there have been reports of reduction in partial thromboplastin time.
- Hepatic function: increases in serum concentrations of bilirubin, transaminases, alkaline phosphatase and lactic dehydrogenase alone or in combination have been reported.
- Central nervous system: headache, paresthesia, convulsions have been reported but a causal relationship with meropenem has not been established.
- Other: oral and vaginal candidiasis.

4.9 Overdose

Accidental overdosage could occur during therapy, particularly in patients with renal impairment. Treatment of overdosage should be symptomatic and supportive. There is no known antidote. In the event of overdosage, meropenem should be discontinued and general supportive treatment given until renal elimination takes place. In normal individuals, rapid renal elimination will occur; in subjects with renal impairment, hemodialysis will remove meropenem and its metabolite.

5. Pharmacological properties

5.1 Pharmacodynamic properties

Meropenem is a carbapenem antibiotic for parenteral use, that is relatively stable to human dehydropeptidase-1 (DHP-1) and therefore does not require the addition of an inhibitor of DHP-1. Meropenem exerts bactericidal activity by inhibiting cell wall synthesis by penetrating the cell wall of most Gram-positive and Gram-negative bacteria to reach penicillin-binding-protein (PBP) targets. Its strongest affinity is toward PBPs 2, 3, and 4 of *Escherichia coli* and *Pseudomonas aeruginosa* and PBPs 1, 2 and 4 of *Staphylococcus aureus*. Bactericidal concentrations are typically one to two times the bacteriostatic concentrations; the exception is *Listeria monocytogenes*, against which lethal activity has not been observed.

Similar to imipenem, the antibacterial action of meropenem is related to binding of the drug to penicillin binding protein (PBPs) of Gram-positive and Gram-negative organisms. The high resistance of meropenem to most bacterial β -lactamases and good penetration of the drug through the outer membrane also contribute significantly to antimicrobial activity. Meropenem may be less of an inducer of β -lactamases than imipenem.

5.2 Pharmacokinetic properties

Absorption

Peak concentration: dependent on dose, renal function and administration technique. The time to peak concentration following intravenous administration is approximately 1 hour (range: 0.5-1.5 hours) after the start of the infusion.

Distribution

Plasma protein binding of meropenem is approximately 2%.

Meropenem achieves concentrations that match or exceed those required to inhibit most susceptible bacteria in most body fluids and tissues including cerebrospinal fluid. Peak concentrations in body fluids and tissues including cerebrospinal fluid. Peak concentrations in body fluids were mostly achieved in 1 hour following intravenous infusion. The volume of distribution is 12 to 20 L.

Metabolism

Extrarenal, 20 to 25% increases up to 50% in patients with creatinine clearance of less than 20 ml/minute. There is one metabolite which is inactive, ICI-213689.

Excretion

Approximately 70% of a meropenem dose administered intravenously is recovered unchanged in the urine over 12 hours. The clearance of meropenem from plasma correlates with the creatinine clearance. There is no accumulation of repeated doses of meropenem 500 mg every 8 hours or 1 gram every 6 hours in patients with normal renal function. Dose adjustments are necessary in patients with renal impairment.

Elimination half-life:

- Adults and children age 2 years and older: 1 hour.
- Children age 3 months to 2 years: 1.5 hours.
- Preterm neonates (27 to 32 weeks gestational age, 21 days mean postnatal age): 3.4 hours.
- Impaired renal function: 3.4 to 20 hours or longer.

5.3 Preclinical safety data

Not applicable.

6. Pharmaceutical particulars**6.1 List of excipients**

Sodium carbonate

6.2 Incompatibilities

Meropenem should not be mixed with or added to other drugs.

6.3 Shelf life

The injection can be used within 24 months from the date of manufacture if kept as recommended.

The reconstituted solution should be used immediately. If not used immediately, it must be stored for no longer than 3 hours at 25°C or 16 hours at 5°C when diluted in water for injection and no longer than 3 hours at 25°C or 15 hours at 5°C when diluted in 0.9% sodium chloride. Meropenem shows lower stability in 5% dextrose compared to other solutions; therefore it should be freshly prepared.

Both the vial before reconstitution and the reconstitution solution should not be frozen.

6.4 Special precautions for storage

Store at temperature below 30°C, protect from light.

6.5 Nature and contents of container

Box containing 10 vials of clear glass thermal coefficient 7 with bromobutyl rubber stopper 20 mm and sealed with aluminium cap 20 mm.

6.6 Special precautions for disposal

Not applicable.

7. Marketing authorization holder

7.1 Manufacturer

PT DEXA Medica
Jl. Jenderal Bambang Utoyo No.138
Palembang-Indonesia

7.2 Product registration holder

Averroes Pharmaceuticals Sdn. Bhd.
3-08-01 & 3-09-01, Block 3,
Presint Alami
Worldwide Business Centre II
Persiaran Akuatik, Section 13
40100 Shah Alam, Selangor Malaysia
Tel: 03-55111433
Fax: 03-55111431

8. Date of revision

07 January 2025